

Bandgap engineering of halide perovskite nanoribbons for high-performance photodetection

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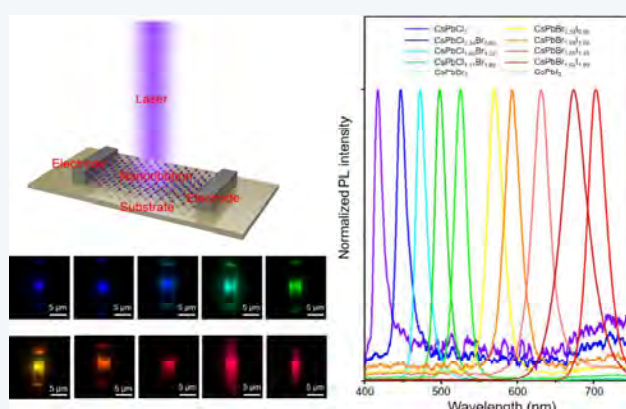
Cite this article: *Nano Research*, 2025, 18, 94907347. <https://doi.org/10.26599/NR.2025.94907347>

ABSTRACT: Bandgap engineering of semiconductor nanowires or nanoribbons (NRs) offers a promising material foundation for multifunctional integrated optoelectronic devices and circuits. Among these materials, all-inorganic halide perovskites have emerged as a leading candidate for next-generation photoelectronic applications due to their outstanding optoelectronic properties. In this work, we report the direct synthesis of high-quality bandgap gradient lead halide perovskite ($\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0-1$)) NRs using a magnetic-pulling source-moving chemical-vapor-deposition (CVD) method. Microstructural characterizations reveal that these as-grown NRs possess high-quality single crystalline structures with continuously tunable compositions. The photoluminescence emissions of these perovskite NRs can be finely tuned across the entire visible spectrum (417–702 nm). Furthermore, photodetectors based on these perovskite NRs demonstrate exceptional photoelectric performance, including a high $I_{\text{ON}}/I_{\text{OFF}}$ ratio (10^4), superior responsivity (37.5 A/W), and remarkable detectivity (2.81×10^{13} Jones). A spatially resolved imaging sensor based on these perovskite NRs is also demonstrated, indicating promising applications in photoelectronic imaging circuits. These bandgap-tunable perovskite NRs provide a versatile materials platform for future integrated devices in electronics and optoelectronics.

KEYWORDS: perovskite nanoribbons, anion exchange process, one-step chemical-vapor-deposition, bandgap engineering, photodetector

1 Introduction

Halide perovskites have attracted significant attention due to their exceptional optoelectronic properties, such as extended carrier



lifetimes, high photoluminescence (PL) quantum yields, and efficient charge transfer [1–3]. These characteristics make halide perovskites ideal material platforms for integrated circuits, including solar cells [4–6], light-emitting diodes (LEDs) [7–9], lasers [10–13], photodetectors [14–18], optical waveguides [19, 20], and more. Traditional semiconductors with single and limited bandgap structures face numerous limitations in the application of integrated optoelectronic devices and circuits. In contrast, the bandgaps of alloy semiconductors can be engineered and tailored to meet diverse application requirements [21, 22]. In this context, the

Received: February 6, 2025; Revised: February 27, 2025

Accepted: March 4, 2025

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anion exchange method has proven to be a simple and versatile approach for achieving precise control over material compositions [23–27]. Specifically, perovskites possess a soft dynamic lattice and low defect formation energy due to their fragile ionic bonds and high vacancy concentrations, which allow ions to diffuse and migrate within the lattice. This makes them ideal materials for bandgap-engineered alloy structures [28–30].

Adjusting the compositions of halide perovskites through ion exchange processes is an effective way to regulate the bandgaps of these semiconductors. For instance, in 2017, Yang et al. reported a three-component heterojunction based on a single nanowire CsPbX_3 ($X = \text{Cl}, \text{Br}, \text{I}$), achieved through partial structure masking and anion exchange [27]. In 2020, Tang et al. constructed a single perovskite alloy nanowire using solid-state anion diffusion technology, which was utilized for a range of continuous wavelength (CW)-tunable lasers [31]. In 2022, Li et al. reported a spatially resolved one-dimensional single Y-CsPbI_3 nanowire through an anion exchange process, which exhibited the formation of a sharp heterojunction at lower temperatures [32]. Moreover, $\text{CsPbCl}_{3-x}\text{Br}_x$ microdisk laser arrays and $\text{CsPb}_x\text{Sn}_{1-x}(\text{Br}_y\text{I}_{1-y})_3$ nanowires have also been successfully realized using the anion exchange method [33, 34]. Despite the significant advantages of perovskite structures in optoelectronics, synthesizing and applying bandgap-continuous gradient perovskite nanoribbons (NRs) with emissions covering the entire visible range remains challenging.

Compared to three-dimensional perovskite bulks and polycrystalline films, perovskite nanoribbons offer a more direct charge transfer path, which benefits charge carrier collection. The larger effective area of NRs enhances photodetectors' performance, providing higher switch ratios, responsivity, and detectivity [35–38]. In this work, we report the successful synthesis of bandgap-continuous tunable all-inorganic halide perovskite NRs using a developed magnetic-pulling source replaced chemical-vapor-deposition approach. Structural characterization of these as-grown NRs reveals a regular stripe structure with tunable compositions and high-quality crystallinity. Spatially resolved photoluminescent spectra and PL mapping of these ribbons show wavelength-continuous tunable emissions from 417 to 702 nm, nearly covering the entire visible range. High-performance photodetectors based on these perovskite NRs were constructed and systematically investigated, demonstrating high responsivity (37.5 A/W), high detectivity (2.81×10^{13} Jones), fast response times (16–17 ms), and high external quantum efficiency (EQE) ($7 \times 10\%$). Moreover, a spatially resolved image sensor based on compositional gradient perovskite NRs was demonstrated to have excellent imaging capability, indicating promising application prospects in future optoelectronic imaging circuits. These results suggest that bandgap-graded perovskite NRs provide an attractive material platform for integrated photonics and optoelectronic devices.

2 Experimental

2.1 Synthesis of pure CsPbCl_3 , CsPbBr_3 , and CsPbI_3 NRs

A magnetic-pulling source-moving CVD system was used to synthesize inorganic halide perovskite NRs, as schematically shown in Fig. S1 in the Electronic Supplementary Material (ESM) and Experimental. Powder sources and reagents were purchased from Alfa Aesar. A horizontal furnace (OTF-1200X) with a quartz tube (inner diameter 45 mm, length 180 cm) was used during the experiment. Before growth, CsBr/PbBr_2 powder sources (molar

ratio = 2:1) were mixed and placed in a quartz boat in the center and upstream of the heating zone A. The product was collected on the Si/SiO_2 substrates (4 mm $\times$ 10 mm) at the deposition area in the center of heating zone B. 0.1 mg/mL Sn solution was dropped onto the substrates as a catalyst before the growth. Before heating, N_2 gas was introduced into the system at a flow rate of 60 sccm for 30 min to purge the air and moisture from the tube. Then, the carrier gas (40 sccm H_2 and 40 sccm N_2) was introduced into the system, and the pressure was maintained at 5.8 Torr. The CsPbX_3 source powder was placed inside the heating zone A with temperature settings of 420 °C (CsPbI_3), 430 °C (CsPbBr_3), and 460 °C (CsPbCl_3). The substrates were placed inside heating zone B with temperature settings of 290 °C (CsPbI_3), 330 °C (CsPbBr_3), and 350 °C (CsPbCl_3) at a rate of 25 °C/min, respectively. After the growth, the temperature was naturally reduced to room temperature.

2.2 Synthesis of alloyed perovskite NRs

A first quartz boat with the mixture of $\text{PbBr}_2/\text{CsBr}$ powder (mole ratio = 1:2, Alfa Aesar, 99.9%) was located at the centre of heating area. A second quartz boat with the mixture of PbX_2/CsX powder ($X = \text{Cl}$ or I , mole ratio = 1:2, Alfa Aesar, 99.9%) was placed away from the heating zone of the furnace before growth. Both quartz boats with powder sources were connected by a quartz rod with sufficient distance between them (Fig. S1 in the ESM). Moreover, the stepper motor (Zolix, PA 100) and the reaction source replacement device were connected by magnets. The stepper motor with the magnet runs 30 cm at a constant speed of 20 cm/min to achieve the replacement of the reaction source. After 60 min growth of the first step, the furnace temperature was reduced to 440 °C ($\text{CsPbCl}_{3-3x}\text{Br}_{3x}$) or 420 °C ($\text{CsPbBr}_{3-3x}\text{I}_{3x}$). The second boat was pushed into the left zone to replace the former boats at a rate of 20 cm/min by a stepping motor. The carrier gas (40 sccm H_2 and 40 sccm N_2) was introduced into the system, and the growth pressure of the nanoribbons was controlled around 5.8 Torr at the entire experimental process. Under high temperatures, these perovskite alloy NRs were composed by controlling the growth time during the second growth process. The relationship between anion exchange time and components was shown in Fig. S1 in the ESM.

2.3 Characterization

The scanning electron microscopy (SEM, Hitachi, SU-8010, Japanese) and transmission electron microscopy (TEM, JEM-F200) were used to investigate the morphology and compositions of the perovskite NRs. Crystal structures were analyzed using the X-ray diffraction (XRD, Bruker D8) system. The optical characterizations were tested by a confocal optical system. The laser beam (375 nm, the power density of laser: 1.4×10^6 mW/cm², spot size: 1.8×10^{-8} cm²) was focused by a microscope lens (Nikon, 100 $\times$) and locally excited at the samples. The PL spectra were recorded by an optical spectrometer (Maya Pro2000), and real-color images were recorded by a CCD camera, as shown in Fig. S2 in the ESM. Before the optical testing, some selected NRs were transferred to the pre-cleaned Si/SiO_2 substrates. The electrodes are completed through a metal evaporation process. The Keithley 2450 was used to investigate the constructed photodetector's current-voltage (I - V) characteristics. The imaging sensors were investigated under a two-dimensional electric displacement table (Zolix-SC300). A signal generator (DG822, Rigol) was used during the optical communication test, providing the output signal to the laser.

3 Results and discussion

Some typical CsPbBr_3 NRs were selected for structural characterizations, as shown in Fig. 1. Figure 1(a) and the inset image show the low- and high-resolution SEM images of the CsPbBr_3 NRs, respectively. More selected SEM images and elemental profiles of CsPbX_3 ribbons are shown in Figs. S3 and S4 in the ESM. The length and the width of the perovskite NRs are ~ 5 – 20 and ~ 2 – 5 μm , respectively, according to the structural investigation. High-resolution SEM images and the energy-dispersive X-ray spectroscopy (EDX) mapping images of a typical pure CsPbBr_3 NR are shown in Fig. 1(b). As can be seen, Cs, Pb, and Br exhibit a uniform distribution along the length of the ribbon. Figure 1(c) shows the EDX spectra at two typical positions (Spot 1 and Spot 2) along a single ribbon, which shows the elements (Cs, Pb, and Br) ratio about 1:1:3, showing well agreement with pure CsPbX_3 perovskite structures. Figure 1(d) shows the EDX line scanning profiles of the same NR along the axial direction, which shows agreement with the two-dimensional (2D) EDX elemental mapping in Fig. 1(b). Notably, Pb exhibits a sudden increase at the tip of the NR, which indicates that the catalyst nanosphere at the tip of the NR is mainly composed of Pb. Meanwhile, the catalysts can also be observed at the tips of some

NRs, as shown in Fig. S4 in the ESM. This is consistent with the previously reported phenomenon of a catalyst-assisted vapor–liquid–solid (V–L–S) growth mechanism [18, 19]. Figure 1(e) shows the XRD patterns of the composition gradient perovskite alloyed NRs, ranging from 10° to 50° . The sharp diffraction peaks in Figs. 1(e) and 1(f) confirm that these composition-tunable CsPbX_3 alloyed nanostructures have high-quality crystalline. The XRD results indicate that CsPbCl_3 , CsPbBr_3 , and CsPbI_3 NRs are indexed to tetragonal, cubic, and orthorhombic phases, respectively, which shows good agreement with the previously reported structures [39, 40]. The strong (100) diffraction peaks of CsPbCl_3 and CsPbBr_3 samples shift significantly towards lower diffraction angles with the increase of Br and I in the corresponding $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ alloys structures, respectively, which indicates that halogen ions completely diffuse with each other during the growth process, forming a random alloy without phase separation [39]. The XRD results of these alloyed CsPbX_3 NRs with different compositions show good agreement with the EDX profiles of the perovskite NRs, as shown in Fig. S5 in the ESM. For example, Fig. 1(g) shows the low-resolution TEM image of a typical CsPbBr_3 NR, which shows a length and width of about 22 and 3 μm , respectively. Figures 1(h) and 1(i) show the HR-TEM images at two typical positions (the red box and blue box in

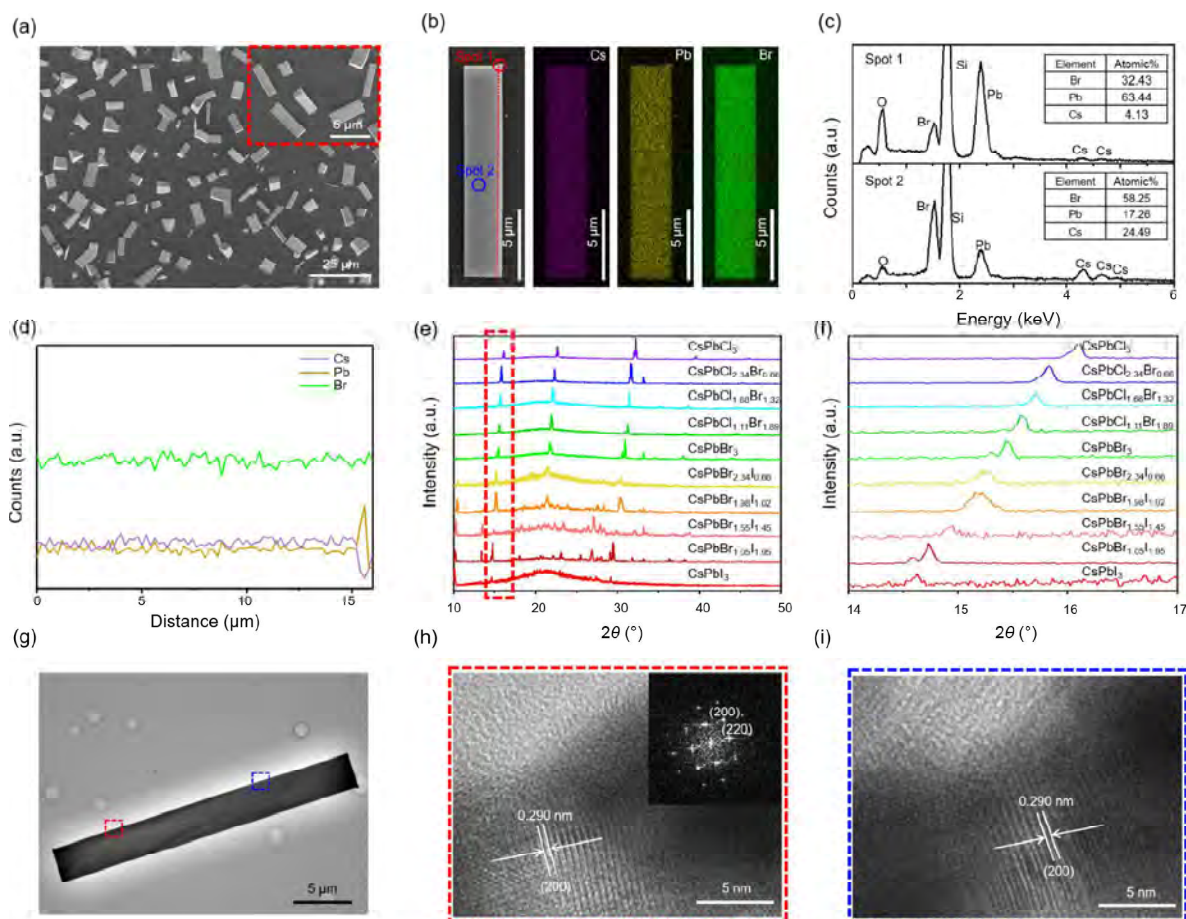


Figure 1 Structural characterizations of a typical CsPbBr_3 NR. (a) Low- and high-resolution (inset) top-view SEM images of CsPbBr_3 perovskite NRs grown on the Si/SiO_2 substrate. (b) SEM image and EDX element mappings of a typical CsPbBr_3 NR. (c) EDX spectra at two typical positions (Spot 1 and Spot 2) of a CsPbBr_3 NR, as indicated in (b). (d) EDX line scans EDX profiles along the axial direction of an NR (dotted line as shown in (b)). (e) XRD patterns of the bandgap gradient $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0$ – 1) NRs ranging from 10° to 50° . (f) Enlarged XRD pattern ranging from 14° to 17° . (g) Low-resolution TEM image of a typical CsPbBr_3 NR. (h) and (i) High-resolution TEM images of two typical positions along the ribbon (inset shows the corresponding FFT patterns of CsPbBr_3 NRs).

Fig. 1(e)) along the length of a typical ribbon. Inter planar spacing of the CsPbBr_3 NR is measured for 0.290 nm, consistent with the (200) plane of the cubic phase CsPbBr_3 [41–43]. The fast Fourier transform (FFT) method obtained the inset, indicating that the CsPbBr_3 NRs are in a well-organized cubic phase, which is in good agreement with the XRD measurements. These results show that the as-grown $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ alloy structures are composition-graded NRs with high-quality crystalline.

The room-temperature optical properties of these NRs were studied by a confocal optical microscopy system, as shown in Fig. S2 in the ESM. PL spectral characterizations, including dark-field real-color images and PL mapping of the typical perovskite NRs, are shown in Fig. 2. Figures 2(a)–2(d) show the top-view optical

photograph and dark-field real-color images of a typical CsPbBr_3 NR, respectively. A 375 nm CW laser is used and focused on the samples as the exciting light. 2D PL mapping (Fig. 2(h)) of the CsPbBr_3 NR shows a uniform emission along the ribbon, which exhibits a perfect belt-like structure. Figures 2(e)–2(g) shows the corresponding PL emission spectra of three typical positions (P_1 – P_3) from the CsPbBr_3 NR, which have emission peaks centered at 525 nm, showing well agreement with the real-color images in Figs. 2(b)–2(d), respectively. Figures 2(i) and 2(m) show the optical images and dark-field real-color images of the pure CsPbCl_3 and CsPbI_3 NRs, respectively. Dark-field images of the CsPbCl_3 and CsPbI_3 NRs are shown in Figs. 2(j) and 2(n); bright blue and red emissions can be observed from the two ribbons under a 375 nm

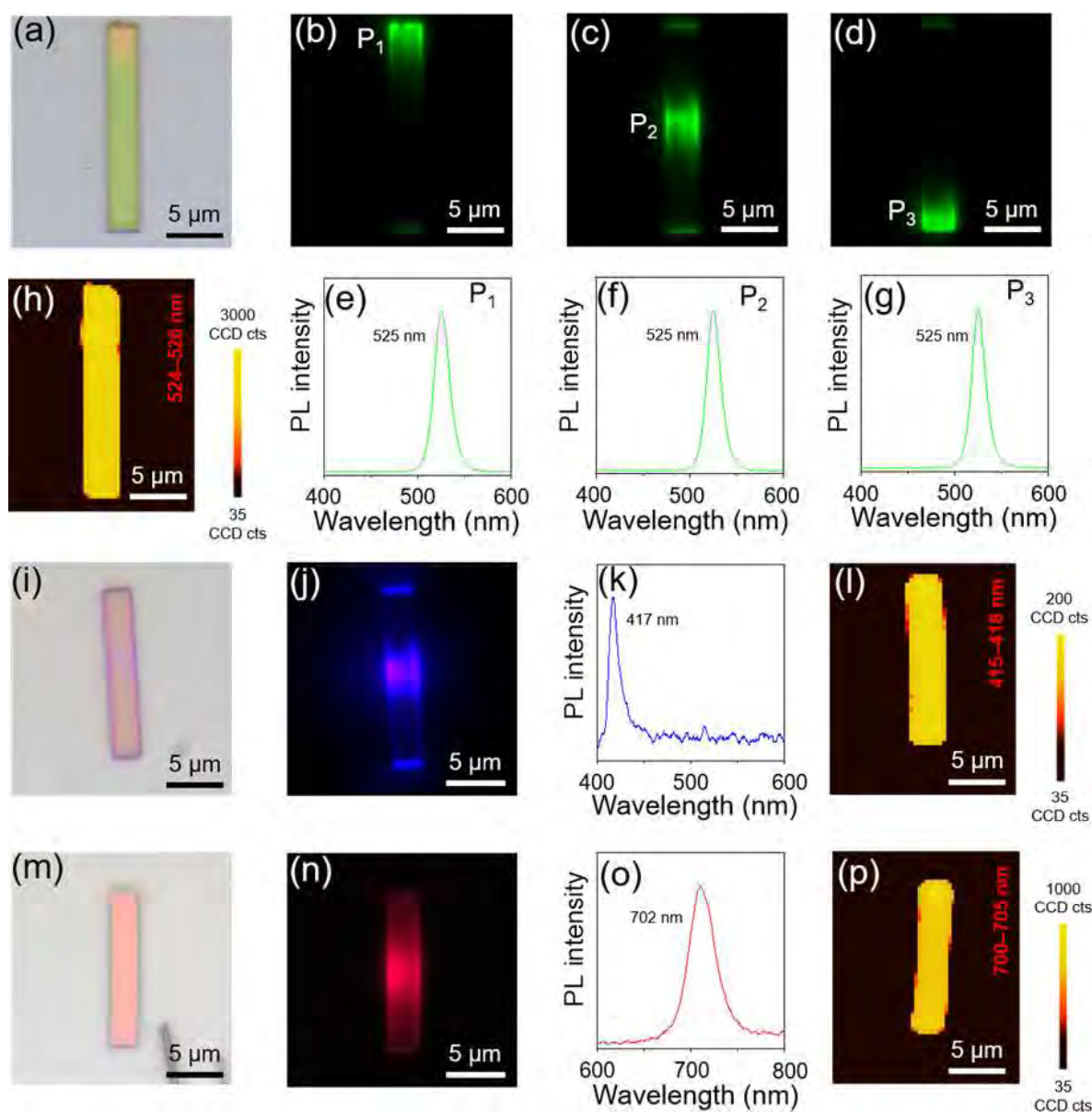


Figure 2 Optical characterizations of the bandgap tunable CsPbX_3 ($X = \text{Cl}, \text{Br}, \text{and I}$) NRs. (a) Optical image of a typical CsPbBr_3 NR. (b)–(d) Dark-field real-color images of the CsPbBr_3 NR excited by a focused 375 nm laser at three typical positions (P_1 – P_3), respectively. (h) 2D PL mapping of a selected CsPbBr_3 NR in (a) (scanning wavelength ranges are 524–526 nm). (e)–(g) The corresponding PL spectra at three typical positions (P_1 – P_3), respectively. (i) and (m)) Optical photographs, (j) and (n)) dark-field images, and (l) and (p)) corresponding PL spectra of the CsPbCl_3 and CsPbI_3 NRs, respectively. ((k) and (o)) 2D PL mapping of the CsPbCl_3 (scanning wavelength ranges are 415–418 nm) and CsPbI_3 (scanning wavelength ranges are 700–705 nm) NRs, respectively.

laser irradiation. Meanwhile, an optical waveguide can be observed at both ends of the NRs. Figures 2(k), 2(l), 2(o), and 2(p) show the PL spectra and 2D PL mapping of CsPbCl₃ and CsPbI₃ NRs, indicating PL emission centered at 417 and 702 nm, respectively. The PL spectra detected from a single NR show that the full width at half maximum (FWHM) of CsPbCl₃, CsPbBr₃, and CsPbI₃ nanowires are about 10, 16, and 31 nm, respectively, suggesting that these pure CsPbX₃ NRs may have good crystalline quality without obvious defects [42–45].

The composition-dependent optical properties of these perovskite alloyed NRs are investigated, as shown in Fig. 3. Figure 3(a) exhibits the dark-field real-color images of the

CsPbCl_{3–3x}Br_{3x} and CsPbBr_{3–3x}I_{3x} ($x = 0–1$) alloyed NRs under a 375 nm laser excitation, which exhibits a color tunable emission from blue to green and red, respectively. The corresponding PL spectra in Fig. 3(b) show that the emission peaks of these alloyed NRs are continuously changed from 417 to 702 nm, nearly covering the entire visible range. The bright luminescence at both ends of the NRs is obviously observed under focused laser excitation, suggesting good waveguide properties of the perovskite NRs. Notably, the FWHM of the PL emission peaks in Fig. 3(b) is about 10–35 nm, suggesting that these as-grown NRs synthesized using a CVD strategy may have high crystallization [46]. Figure 3(c) shows the relationship between the component x and the bandgap values

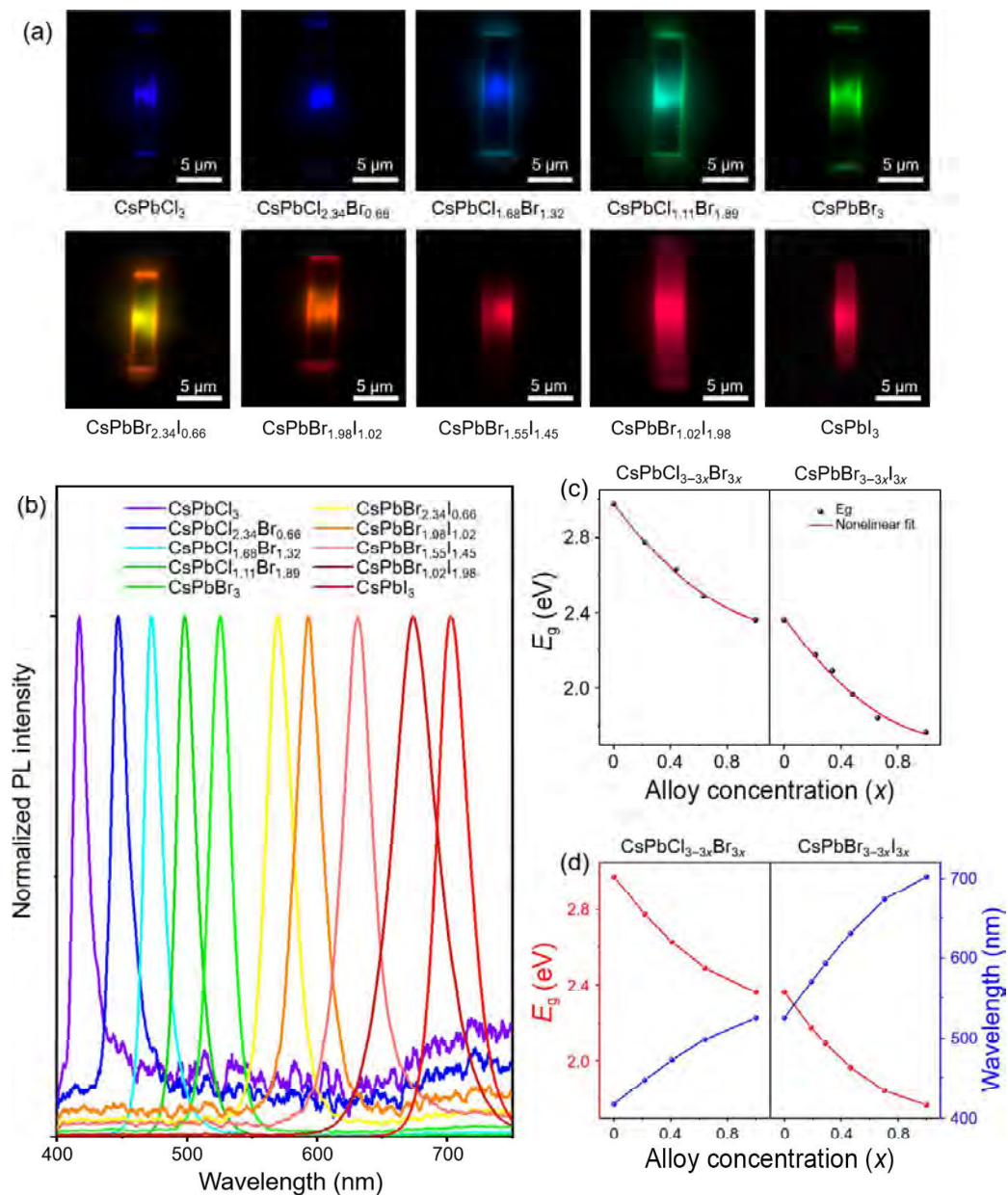


Figure 3 Optical properties of bandgap gradient perovskite CsPbX₃ alloy NRs. (a) Dark-field real-color images and (b) corresponding PL spectra of bandgap gradient CsPbCl_{3–3x}Br_{3x} and CsPbBr_{3–3x}I_{3x} ($x = 0–1$) alloy NRs with different components excited by a 375 nm laser beam. The scale bar is 5 μm. (c) The relationship between bandgaps and compositions of Br ratio in alloy NRs obtained by the EDX profiles. The black square represents the experimental data E_g , and the solid red line represents the fitting line. (d) The relationship between emission wavelength and bandgap of the CsPbCl_{3–3x}Br_{3x} and CsPbBr_{3–3x}I_{3x} ($x = 0–1$) alloy NRs. The red and blue circles represent the bandgaps of CsPbX₃ NRs and the wavelength of the emission peaks, respectively.

of the $\text{CsPbCl}_{3-x}\text{Br}_{3-x}$ and $\text{CsPbBr}_{3-x}\text{I}_{3-x}$ alloyed ribbons obtained through the EDX profiles (Fig. S3 in the ESM and Experimental). The relationship is shown as follows

$$E_g(\text{A}_x\text{B}_{1-x}) = xE_g(\text{A}) + (1-x)E_g(\text{B}) - bx(1-x) \quad (1)$$

For Cl/Br and Br/I alloys, the compositional dependence of the bandgap can be expressed as

$$E_g(\text{CsPbCl}_{3-3x}\text{Br}_{3x}) = xE_g(\text{CsPbBr}_3) + (1-x)E_g(\text{CsPbCl}_3) - bx(1-x) \quad (2)$$

$$E_g(\text{CsPbBr}_{3-3x}\text{I}_{3x}) = xE_g(\text{CsPbI}_3) + (1-x)E_g(\text{CsPbBr}_3) - bx(1-x) \quad (3)$$

where x represents the proportion of Br element components, and b represents the bending parameter of the curve. $E_g = 2.974$, 2.362, and 1.767 eV represent the bandgaps of CsPbCl_3 , CsPbBr_3 , and CsPbI_3 , respectively. Finally, the relationship curve between the

bandgap of CsPbX_3 and Br element content is obtained. According to the calculation results, the values of $b_{\text{Cl/Br}}$ and $b_{\text{Br/I}}$ are 0.37 and 0.42, respectively, nearly equal to the previously reported results [21, 47]. Figure 3(d) shows the relationship between the component Br, emission wavelength, and bandgaps of different alloy NRs. The solid lines represent the best fit of component-related bandgaps using nonlinear fitting. As can be seen, the EDX results are consistent with the calculated bandgap values, confirming the accurate regulation of perovskite components. These results indicate that these bandgap gradient perovskite alloy NRs show excellent luminescent performance with no obvious defect emissions, which agrees well with the structural characterizations in Fig. 2.

The above discussions clearly demonstrate the successful realization of the bandgap gradient CsPbX_3 alloyed NRs. To demonstrate their potential optoelectronic applications, photodetectors based on these perovskite NRs were constructed on a SiO_2 substrate, and their photoconductive properties were systematically investigated. Figure 4(a) shows a schematic diagram

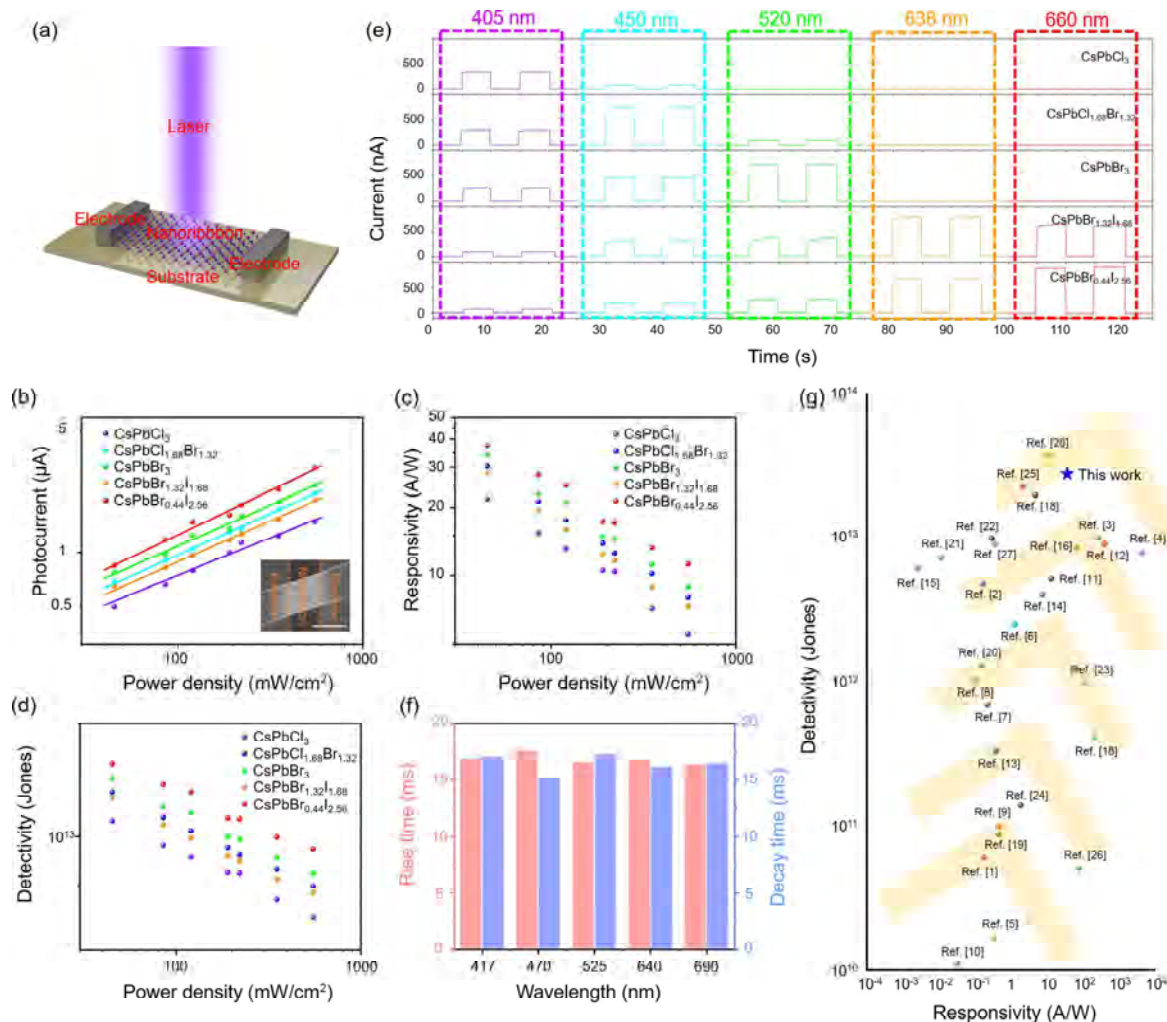


Figure 4 Photoelectrical properties of bandgap gradient $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0-1$) NRs based photodetectors. (a) Schematic diagram of a photodetector based on a CsPbX_3 NR. (b)–(d) Light intensity-dependent photocurrent, responsivity, and detectivity of the different composition NR-based photodetectors. Inset shows the SEM image of the as-fabricated NRs PD, the scale bar is 6 μm . (e) The I - T curves of $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0-1$) NRs-based photodetectors under various light illumination (405, 450, 520, 638, and 660 nm, power density = 45 mW/cm^2). (f) Rise and decay time of $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0-1$) NRs-based photodetectors under 405, 450, 520, 638, and 660 nm illumination. (g) Comparison of the perovskite NR-based photodetectors' responsivity and detectivity with reported literature. Detailed performance parameters of these devices are shown in Table S1 in the ESM and Experimental.

of a photodetector using an individual perovskite NR. I - V , I - T characteristics and stability testing of all devices are shown in Figs. S6–S10 in the ESM. Each device's $I_{\text{ON}}/I_{\text{OFF}}$ ratio can be calculated based on the measured light and dark current values. After calculation, the $I_{\text{ON}}/I_{\text{OFF}}$ ratios of all devices reached 10^4 or above, indicating a high performance of the devices. The SEM image of the device is shown in the inset of Fig. 4(b), in which the electrodes with a gap of 5 μm are thermally deposited on both ends of the ribbon, and the thickness of the electrode is controlled at 200 nm. Detailed experimental results are shown in the Experiment sections.

It is also noted that the laser beam with a wavelength of 660, 638, 520, 450, and 405 nm was used as the light sources for the photodetection measurement of $\text{CsPbBr}_{0.44}\text{I}_{2.56}$, $\text{CsPbBr}_{1.32}\text{I}_{1.68}$, CsPbBr_3 , $\text{CsPbCl}_{1.68}\text{Br}_{1.32}$, and CsPbCl_3 NR photodetectors, respectively. The responsivity (R) and detectivity (D^*) are two critical parameters to assess the performance of photodetectors, which can be calculated by the equations as follows [48, 49]

$$R = \frac{I_{\text{ph}}}{PA} \quad (4)$$

$$D^* = R \sqrt{\frac{A}{2eI_{\text{dark}}}} \quad (5)$$

where I_{ph} is the difference between the photocurrent and dark current, $P = 45 \text{ mW/cm}^2$ is the incident light power density illuminated at the periodic superstructures, A is an effective irradiated area on the device, and e is the electronic charge. Both R and D^* increase dramatically with a decrease in the light intensity. After measurement, the A values of all devices are concentrated around $4 \times 10^{-7} \text{ cm}^2$. Figures 4(b)–4(d) depict the values of I_{ph} , R , and D^* at different light intensities. It shows that both R and D^* increase dramatically with the decreasing light intensity. Notably, photocurrent and light intensity of the NRs-based photodetectors are observed as a sublinear relationship, which may originate from the complex processes of electron-hole generation, trapping, and recombination in the nanostructures [17, 50]. Under a lower light intensity of 45 mW/cm^2 , the R values of the perovskite alloyed NRs-based photodetectors can reach 37.5, 28.6, 34.4, 30.4, and 21.8 A/W, respectively. At the same time, the corresponding D^* values are up to 2.81×10^{13} , 1.76×10^{13} , 2.29×10^{13} , 1.88×10^{13} , and 1.25×10^{13} Jones, the I_{dark} of different nanoribbons at 2 V bias is 1.33 pA ($\text{CsPbBr}_{0.44}\text{I}_{2.56}$), 1.49 pA ($\text{CsPbBr}_{1.32}\text{I}_{1.68}$), 1.33 pA (CsPbBr_3), 1.61 pA ($\text{CsPbCl}_{1.68}\text{Br}_{1.32}$), and 1.74 pA (CsPbCl_3), respectively. By fitting the relationship between I_{ph} and P ($I_{\text{ph}} \propto P^k$, k represents the correlation coefficient), k is evaluated as 0.5, comparable with the previously reported photodetectors [51]. Figure 4(e) shows the comparison investigation of the τ_r and τ_d of five typical perovskite alloy devices with various bandgaps. According to the results, the rise and decay times of all devices are concentrated at about 16–17 ms, which exhibits a fast response speed of the millisecond magnitude. Figure 4(f) shows the photoresponse properties of five proposed perovskite devices under various light illuminations at $V = 2 \text{ V}$. All devices show significant photoelectrical response to light near their absorption edge. The photocurrents of CsPbCl_3 , $\text{CsPbCl}_{1.68}\text{Br}_{1.32}$, and CsPbBr_3 NR devices are relatively lower than that of the $\text{CsPbBr}_{0.44}\text{I}_{2.56}$ and $\text{CsPbBr}_{1.32}\text{I}_{1.68}$ NR devices under the 638 and 660 nm laser illumination, which may because the energy of the external laser is much smaller than the bandgap of the three previous alloy materials. Additionally, the performances of photodetectors based on most perovskite structures reported to

date are shown in Fig. 4(g) and Table S1 in the ESM and Experimental, indicating that our device performance has significant advantages compared with the reported results. The EQE, which is the number of electron-hole pairs produced by one absorbed photon, is a critical parameter for photodetectors and can be expressed as

$$\text{EQE} = \frac{hcR}{e\lambda} \quad (6)$$

The corresponding calculated EQE are $7.05 \times 10^3\%$, $5.54 \times 10^3\%$, $7.89 \times 10^3\%$, $7.85 \times 10^3\%$, and $6.48 \times 10^3\%$, respectively for photodetectors based on NRs.

In consideration of the high photosensitivity of halide perovskites in the visible light region, high-performance photodetectors may have good potential in image-sensing applications. Figure 5(a) shows the schematic diagram of the imaging sensing system. A 405 nm laser was used as the light source and placed before the image mask. The mask can be controlled by the stepper motor and moved on the X - Y plane. The real-time spatially resolved light response can be recorded, forming an “LIGHT” image, as shown in Fig. 5(b). When a 405 nm light scans along the image mask “LIGHT” along the Y direction with represented X -direction pixel sequences, the time-resolved currents of the perovskite NR-based photodetectors are displayed in Figs. 5(c) and 5(d), confirming the high stability and photosensitivity of the imaging system. Figure 5(e) shows the schematic of the CsPbCl_3 -based device in an optical communication system. Using a signal generator, voltage signals representing the ASCII code “LIGHT” were sent to the laser, as shown in Fig. 5(f), the laser's emission can be controlled. After receiving the light signal, the photodetector generated a current signal representing the ASCII code “LIGHT” as shown in Fig. 5(g). This signal could be further output and decoded to the information “LIGHT” there by realizing the optical communication. These impressive results of the photodetectors and imaging sensors may show promising future applications in the integrated optoelectronic field.

4 Conclusions

In summary, we developed a source-moving CVD approach to synthesize composition tunable $\text{CsPbCl}_{3-3x}\text{Br}_{3x}$ and $\text{CsPbBr}_{3-3x}\text{I}_{3x}$ ($x = 0$ –1) alloy NRs. Structural characterization of these perovskite structures indicates that the obtained alloyed ribbons have high-quality crystalline. The bandgap of these compositional gradient perovskite alloy NRs can be continuously tuned from 1.7–2.9 eV, with tunable emissions of 417–702 nm, nearly covering the entire visible light. Moreover, photodetectors based on these NRs show superior detection performances, including high responsivity (37.5 A/W), high detectivity (2.81×10^{13} Jones), fast response time (16–17 ms), as well as high EQE ($7 \times 10^3\%$). A high-resolution imaging sensor based on the perovskite NRs has also been demonstrated to identify its good imaging capability. These achievements represent significant advantages in the controllable growth of bandgap tunable perovskite NRs, which may act as effective emission systems and have potential applications in high-performance integrated nanophotonic devices.

Electronic Supplementary Material: Supplementary material (CVD setup, SEM image, EDX spectra, characterization of electrical properties of photodetectors, schematic diagram of confocal

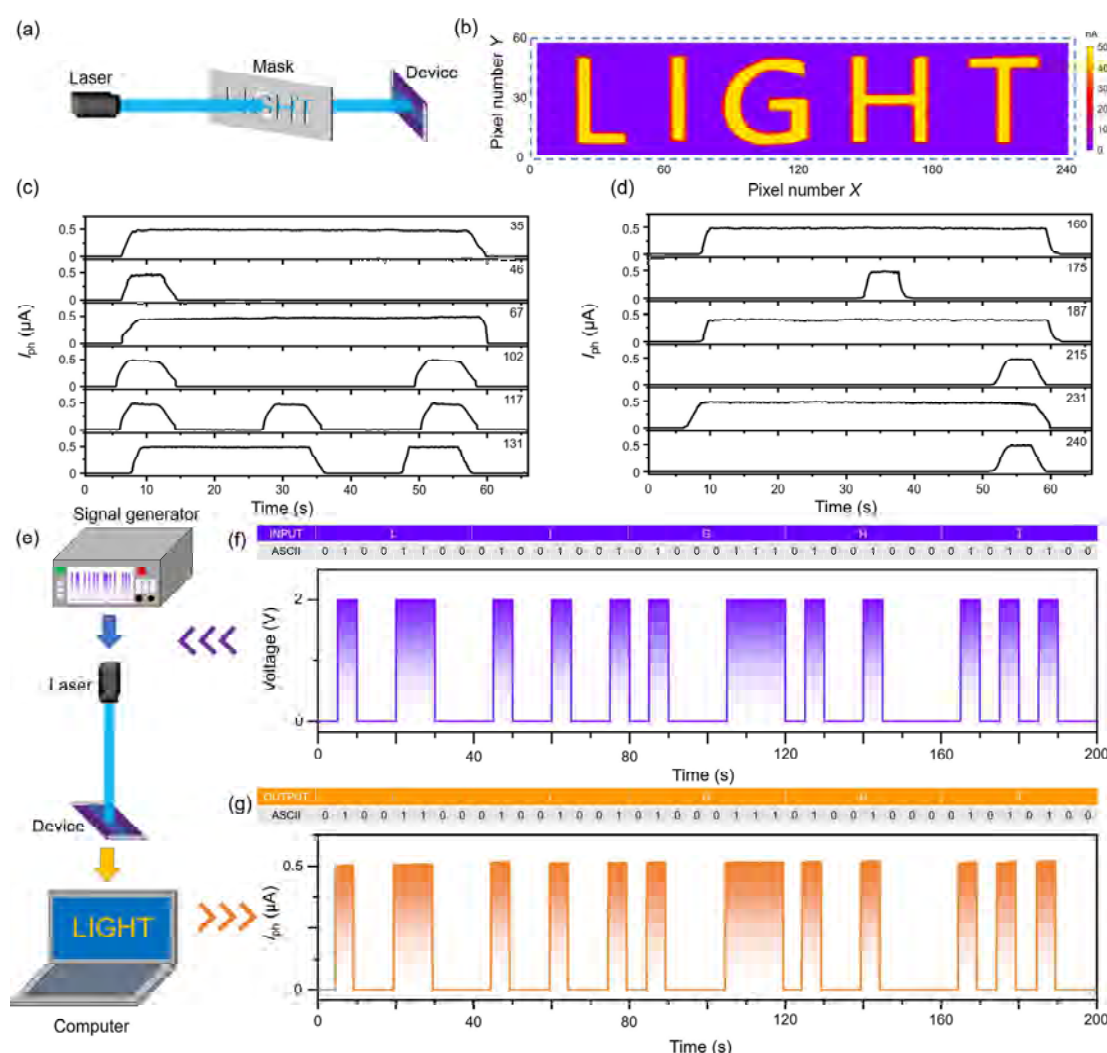


Figure 5 Imaging and optical communication application of the CsPbCl₃-based photodetectors. (a) Schematic diagram of the imaging sensing system. (b) Corresponding two-dimensional photocurrent mapping of image “LIGHT” under a 405 nm laser illumination. (c) and (d) The time-dependent current of “LIGHT” under a 405 nm laser scans along the Y-direction with different X-pixel points. The numbers on the right side indicate the pixel sequences along the X direction. (e) Schematic illustration of the CsPbCl₃-based device for an optical communication system. (f) The input signals of the ASCII code “LIGHT”. (g) The output photocurrent signals of the ASCII code “LIGHT” by the CsPbCl₃-based device.

microscopy system) is available in the online version of this article at <https://doi.org/10.26599/NR.2025.94907347>.

Data availability

All data needed to support the conclusions in the paper are presented in the manuscript or the Electronic Supplementary Material. Additional data related to this paper may be requested from the corresponding author upon request.

Acknowledgements

The authors are grateful to the National Natural Science Foundation of China (Nos. 52373246, 62127817, and U23A20380), the National Key Research and Development Program of China (No. 2022YFA1404201), Science and Technology Major Special Project of Shanxi Province (No. 202201010101005), the Shanxi Province Graduate Innovation Project (No. RC2400005577), and the Shanxi Basic Research Program Project (No. 20210302123128)

for financial support. The authors acknowledge the assistance of the Instrumental Analysis Center at Taiyuan University of Technology.

Declaration of competing interest

All the contributing authors report no conflict of interests in this work.

Author contribution statement

P. F. G. and L. T. X. conceived and designed the research. J. S. W. and X. S. synthesized the perovskite nanoribbons. J. S. W., X. Y. L., and J. C. H. performed device fabrication and electrical characterizations. J. S. W. and X. S. contributed to optical characterizations or analysis. J. S. W., P. F. G., and L. T. X. wrote the related discussions. Q. H. L. participated in the SEM and TEM analysis. P. F. G., J. S. W., and L. T. X. co-wrote the manuscript with all the authors' input. All authors discussed the results and commented on the manuscript.

Use of AI statement

None.

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